Harnam Singh

Software Engineer | Full Stack Developer
Kolkata, India | +91 80172 19644 | harnamsinghsandhu04@gmail.com | LinkedIn | GitHub | Portfolio

SKILLS

Languages: JavaScript, TypeScript, SQL
Backend: Node.js, NestJS, REST APIs, TypeORM
Databases: PostgreSQL, Firestore
Frontend: React, Next.js (SSG, ISR), Tailwind, MUI, SEO, Responsive Design
State Management & Data Fetching: React Query, Zustand, React Context
Security: JWT Authentication, Web Crypto API, AES-256, PBKDF2
Tools: Git, GitHub, Axios

EXPERIENCE

Software Developer Intern - *SigiloTech Pvt Ltd* — View Certificate

- Developed backend APIs and business logic for a school operations platform using NestJS, TypeScript, TypeORM and PostgreSQL, covering child management, classroom assignments, scheduling and capacity management.
- Implemented child journey forecasting, age-group progression and roster scheduling, handling date-based transitions, early moves and temporary schedule overrides.
- Built room capacity calculations and validation using age-group ratios, room configurations, child schedules and predicted transitions; reduced redundant database calls by reusing data across calculations.
- Developed React-based frontend workflows for roster and classroom management, integrating backend APIs using React Query and Material UI.
- Independently developed the complete frontend of the company's admin dashboard for client, job, blog and applicant management, integrating backend CRUD APIs.
- Developed and optimised the company's public-facing website using Next.js, React and Tailwind CSS, implementing responsive components, SSG/ISR, image and route-loading optimisations, and technical SEO.

PROJECTS

Personal Secret Manager (Next.js, TypeScript, Web Crypto API) — Live | GitHub

- Built a credential manager using AES-256 encryption and PBKDF2-based key derivation.
- Designed modular encryption and storage layers to prevent plaintext credential storage in localStorage.

NutriLens — Food Explorer (Next.js, React Query, Tailwind CSS) — Live | GitHub

- Built a food product search application using OpenFoodFacts API, supporting name/barcode search and dynamic routing.
- Implemented infinite scrolling, filtering, sorting and cart management using React Query and React Context.

iOS-Style Calculator Web App (Next.js, Tailwind CSS) — Live | GitHub

- Built a responsive calculator supporting 10+ operations, chained calculations and percentage handling.
- Implemented a custom two-stack expression evaluator with operator precedence and O(n) evaluation.

CERTIFICATIONS & ACHIEVEMENTS

NPTEL - Computer Graphics (IIT Guwahati) [Top 1%] | View Certificate
JavaScript Algorithms and Data Structures — freeCodeCamp | View Certificate

EDUCATION

Master of Computer Applications (MCA), *IGNOU* — JAN 2026 - DEC 2027 (ODL)
B.Sc. Mathematics (Honours), *University of Calcutta* — 2022-25 | CGPA: 6.18/10